Structural Bioinformatic p1

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##Introduction to the RCSB Protein Data Bank (PDB)

PDB statistics

```
data <- read.csv("Data Export Summary.csv", stringsAsFactors = FALSE)

#View(data)
```

```
pdb <- read.csv("Data Export Summary.csv")

pdb
```

```
           Molecular.Type   X.ray     EM    NMR Integrative Multiple.methods
1          Protein (only) 178,795 21,825 12,773         343              226
2 Protein/Oligosaccharide  10,363  3,564     34           8               11
3              Protein/NA   9,106  6,335    287          24                7
4     Nucleic acid (only)   3,132    221  1,566           3               15
5                   Other     175     25     33           4                0
6  Oligosaccharide (only)      11      0      6           0                1
  Neutron Other   Total
1      84    32 214,078
2       1     0  13,981
3       0     0  15,759
4       3     1   4,941
5       0     0     237
6       0     4      22
```

Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

```r
pdb_num <- pdb
pdb_num[-1] <- lapply(pdb_num[-1], function(x) as.numeric(gsub(",", "", x)))

total <- sum(pdb_num$Total)

xray <- sum(pdb_num$X.ray)

em <- sum(pdb_num$EM)

xray_percent <- xray / total * 100
em_percent <- em / total * 100

xray_percent
```

```
[1] 80.95077
```

```r
em_percent
```

```
[1] 12.83843
```

> Q2: What proportion of structures in the PDB are protein?

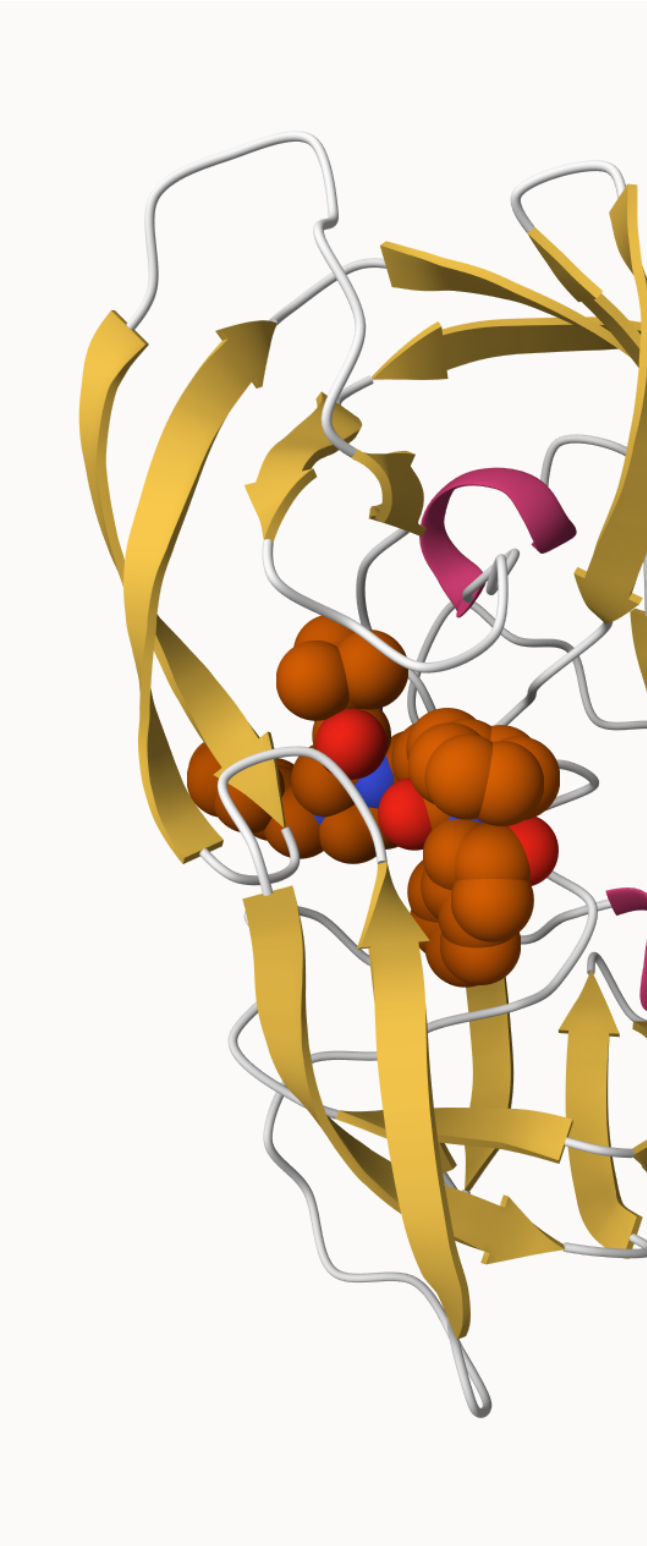
```r
protein_rows <- pdb_num$Molecular.Type %in% c("Protein (only)", "Protein/Oligosaccharide", "P

protein_total <- sum(pdb_num$Total[protein_rows])

protein_percent <- protein_total / total * 100

protein_percent
```

```
[1] 97.9118
```

> Q3: Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

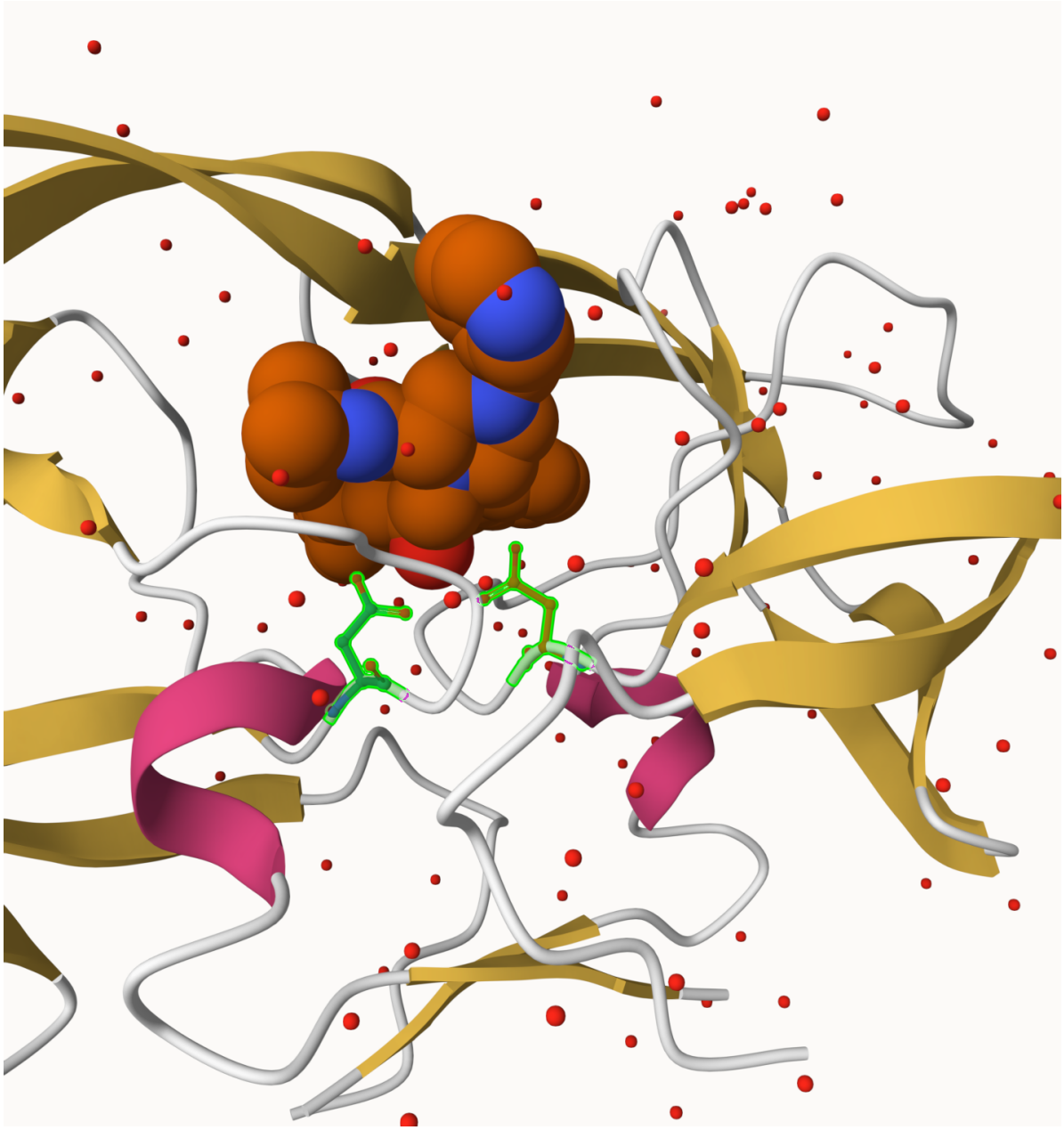
Searching "HIV-1 protease" on the RCSB PDB homepage returns 1,173 structures.

##Visualizing the HIV-1 protease structure



Using Mol Overall structure of HIV-1 protease (PDB: 1HSG)

Close up of the HIV-1 protease active site highlighting the two catalytic Asp25 residues (one from each chain) shown in green, along with the bound ligand and nearby water molecules.



> Q4: Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

In the structure, each water molecule is shown as one atom because X-ray structures usually model only the oxygen atom of water (the hydrogens are not resolved well), so the water

appears as a single sphere/point.

> Q5: There is a critical "conserved" water molecule in the binding site. Can you identify this water molecule? What residue number does this water molecule have

The conserved water molecule in the HIV-1 protease active site is associated with the catalytic residue Asp25 (one in each chain).

> Q6: Generate and save a figure clearly showing the two distinct chains of HIV-protease along with the ligand. You might also consider showing the catalytic residues ASP 25 in each chain and the critical water (we recommend "Ball & Stick" for these side-chains). Add this figure to your Quarto document.

see image above.

##Introduction to Bio3D in R

```
library(bio3d)
```

Reading PDB file data into R

```
pdb <- read.pdb("1hsg")
```

```
  Note: Accessing on-line PDB file
```

```
pdb
```

```
 Call:  read.pdb(file = "1hsg")

   Total Models#: 1
     Total Atoms#: 1686,  XYZs#: 5058  Chains#: 2  (values: A B)

     Protein Atoms#: 1514  (residues/Calpha atoms#: 198)
     Nucleic acid Atoms#: 0  (residues/phosphate atoms#: 0)

     Non-protein/nucleic Atoms#: 172  (residues: 128)
     Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]

   Protein sequence:
      PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
      QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
```

```
      ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
      VNIIGRNLLTQIGCTLNF

+ attr: atom, xyz, seqres, helix, sheet,
        calpha, remark, call
```

Q7: How many amino acid residues are there in this pdb object?

198

Q8: Name one of the two non-protein residues?

HOH and MK1

Q9: How many protein chains are in this structure?

2

```r
attributes(pdb)
```

```
$names
[1] "atom"   "xyz"    "seqres" "helix"  "sheet"  "calpha" "remark" "call"

$class
[1] "pdb" "sse"
```

```r
head(pdb$atom)
```

```
  type eleno elety  alt resid chain resno insert      x      y     z o     b
1 ATOM     1     N <NA>   PRO     A     1   <NA> 29.361 39.686 5.862 1 38.10
2 ATOM     2    CA <NA>   PRO     A     1   <NA> 30.307 38.663 5.319 1 40.62
3 ATOM     3     C <NA>   PRO     A     1   <NA> 29.760 38.071 4.022 1 42.64
4 ATOM     4     O <NA>   PRO     A     1   <NA> 28.600 38.302 3.676 1 43.40
5 ATOM     5    CB <NA>   PRO     A     1   <NA> 30.508 37.541 6.342 1 37.87
6 ATOM     6    CG <NA>   PRO     A     1   <NA> 29.296 37.591 7.162 1 38.40
  segid elesy charge
1  <NA>     N   <NA>
2  <NA>     C   <NA>
3  <NA>     C   <NA>
4  <NA>     O   <NA>
5  <NA>     C   <NA>
6  <NA>     C   <NA>
```

##Quick PDB visualization in R

```
pak::pak("bioboot/bio3dview")
```

```
! Using bundled GitHub PAT. Please add your own PAT using `gitcreds::gitcreds_set()`.

  Loading metadata database

  Loading metadata database ... done





  No downloads are needed

  1 pkg + 40 deps: kept 30 [4.5s]
```

##Predicting functional motions of a single structure

```
adk <- read.pdb("6s36")
```

```
   Note: Accessing on-line PDB file
    PDB has ALT records, taking A only, rm.alt=TRUE
```

```
adk
```

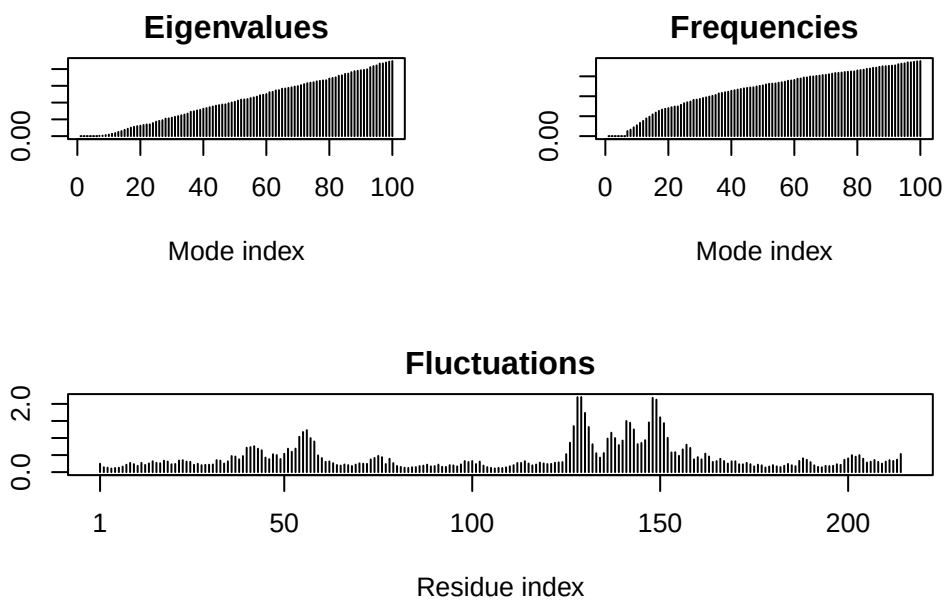
```
 Call:  read.pdb(file = "6s36")

   Total Models#: 1
     Total Atoms#: 1898,  XYZs#: 5694  Chains#: 1  (values: A)

     Protein Atoms#: 1654  (residues/Calpha atoms#: 214)
     Nucleic acid Atoms#: 0  (residues/phosphate atoms#: 0)

     Non-protein/nucleic Atoms#: 244  (residues: 244)
     Non-protein/nucleic resid values: [ CL (3), HOH (238), MG (2), NA (1) ]
```

```
   Protein sequence:
      MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLVT
      DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDKI
      VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
      YYSKEAEAGNTKYAKVDGTKPVAEVRADLEKILG

+ attr: atom, xyz, seqres, helix, sheet,
        calpha, remark, call
```

```r
m <- nma(adk)
```

```
 Building Hessian...        Done in 0.013 seconds.
 Diagonalizing Hessian...   Done in 0.281 seconds.
```

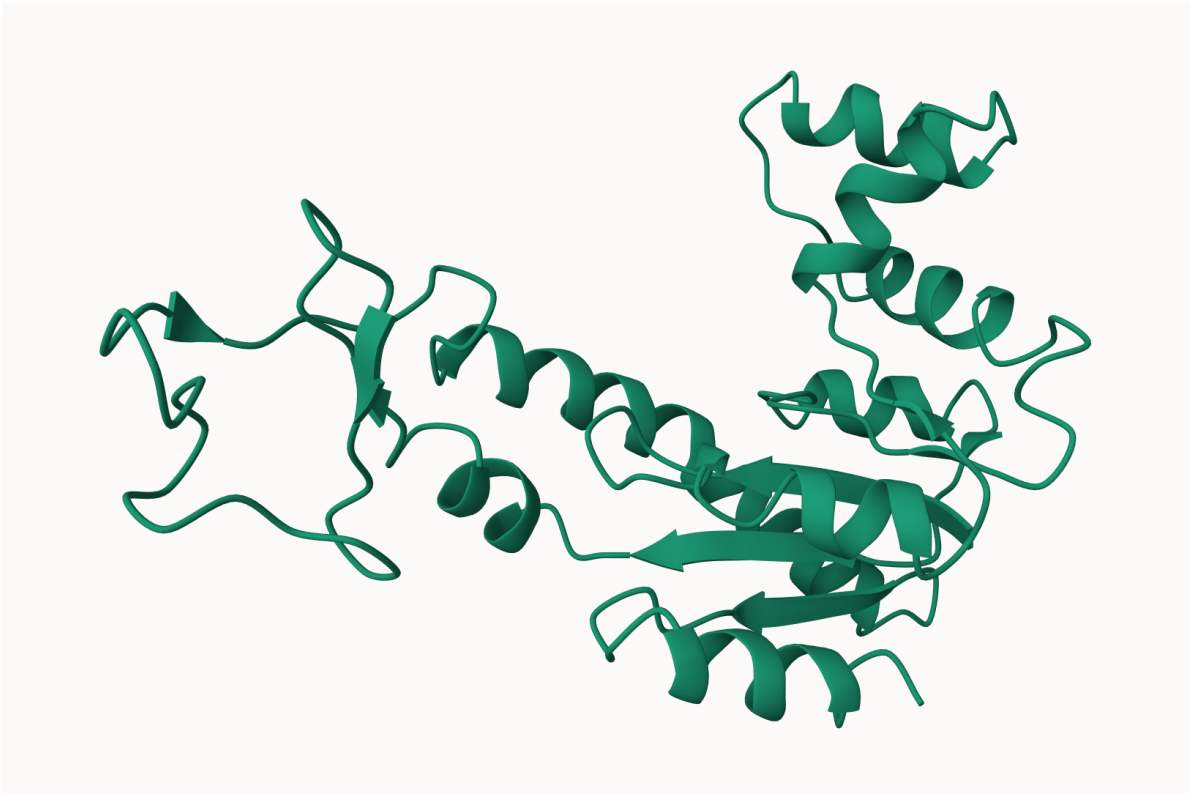
```r
plot(m)
```



Normal mode analysis shows that certain regions of adenylate kinase have much higher fluctuations, indicating these residues are more flexible and likely involved in functional motion.

```r
mktrj(m, file="adk_m7.pdb")
```

```r
knitr::include_graphics("ADK_M7.PDB.png")
```



```r
#view.nma(m, pdb=adk)
```

##Comparative structure analysis of Adenylate Kinase

```r
library(bio3d)
#library(bio3dview)
library(NGLVieweR)
```

Q10. Which of the packages above is found only on BioConductor and not CRAN?

msa

Q11. Which of the above packages is not found on BioConductor or CRAN?

bio3dbiew

Q12. True or False? Functions from the pak package can be used to install packages from GitHub and BitBucket?

True

##Search and retrieve ADK structures

```
library(bio3d)
aa <- get.seq("1ake_A")
```

```
Warning in get.seq("1ake_A"): Removing existing file: seqs.fasta


Fetching... Please wait. Done.
```

```
aa
```

```
              1        .         .         .         .         .         60
pdb|1AKE|A    MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLVT
              1        .         .         .         .         .         60

             61        .         .         .         .         .         120
pdb|1AKE|A    DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDRI
             61        .         .         .         .         .         120

            121        .         .         .         .         .         180
pdb|1AKE|A    VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
            121        .         .         .         .         .         180

            181        .         .         .   214
pdb|1AKE|A    YYSKEAEAGNTKYAKVDGTKPVAEVRADLEKILG
            181        .         .         .   214

Call:
  read.fasta(file = outfile)

Class:
  fasta

Alignment dimensions:
  1 sequence rows; 214 position columns (214 non-gap, 0 gap)

+ attr: id, ali, call
```

Q13. How many amino acids are in this sequence, i.e. how long is this sequence?

Alignment dimensions: 1 sequence rows; 214 position columns.

```
hits <- NULL
hits$pdb.id <- c('1AKE_A','6S36_A','6RZE_A','3HPR_A','1E4V_A','5EJE_A','1E4Y_A','3X2S_A','6HA
```

```
head (hits$pdb.id)
```

```
[1] "1AKE_A" "6S36_A" "6RZE_A" "3HPR_A" "1E4V_A" "5EJE_A"
```

```
files <- get.pdb(hits$pdb.id, path="pdbs", split=TRUE, gzip=TRUE)
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1AKE.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6S36.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6RZE.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3HPR.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4V.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/5EJE.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4Y.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3X2S.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAP.pdb.gz exists. Skipping download
```

```
  |
  |                                                                      |   0%
  |
  |========                                                              |  11%
  |
  |================                                                      |  22%
  |
  |========================                                              |  33%
  |
  |================================                                      |  44%
  |
  |========================================                              |  56%
  |
  |================================================                      |  67%
  |
  |========================================================              |  78%
  |
  |==================================================================    |  89%
  |
  |======================================================================| 100%
```

##Align and superpose structures We need one package from bioConductor. to set this up we need to first install a package called *biocManager** from CRAN.

Now we can use the "instal()" function from this package liek this: `BiocManager::install("msa")`

```
pdbs <- pdbaln(files, fit=TRUE, exefile="msa")
```

```
Reading PDB files:
pdbs/split_chain/1AKE_A.pdb
pdbs/split_chain/6S36_A.pdb
pdbs/split_chain/6RZE_A.pdb
pdbs/split_chain/3HPR_A.pdb
pdbs/split_chain/1E4V_A.pdb
pdbs/split_chain/5EJE_A.pdb
pdbs/split_chain/1E4Y_A.pdb
pdbs/split_chain/3X2S_A.pdb
pdbs/split_chain/6HAP_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
```

```
.   PDB has ALT records, taking A only, rm.alt=TRUE
..   PDB has ALT records, taking A only, rm.alt=TRUE
....

Extracting sequences

pdb/seq: 1   name: pdbs/split_chain/1AKE_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 2   name: pdbs/split_chain/6S36_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 3   name: pdbs/split_chain/6RZE_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 4   name: pdbs/split_chain/3HPR_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 5   name: pdbs/split_chain/1E4V_A.pdb
pdb/seq: 6   name: pdbs/split_chain/5EJE_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 7   name: pdbs/split_chain/1E4Y_A.pdb
pdb/seq: 8   name: pdbs/split_chain/3X2S_A.pdb
pdb/seq: 9   name: pdbs/split_chain/6HAP_A.pdb
```

let's have a wee peak at out strucutres after "fitting" or superposing:

```
#library(bio3dview)

#view.pdbs(pdbs)
```

We can run functions like `rmsd()`, `rmsf()` and the best `pca()`

```
pc.xray <- pca(pdbs)
```

Figure 7: Schematic representation of alignment. Grey regions depict aligned residues, while white depict gap regions. The red bar at the top depict sequence conservation.

##Annotate collected PDB structures

```
ids <- basename.pdb(pdbs$id)

anno <- pdb.annotate(ids)
unique(anno$source)
```

```
[1] "Escherichia coli"
[2] "Escherichia coli K-12"
[3] "Escherichia coli O139:H28 str. E24377A"
[4] "Escherichia coli str. K-12 substr. MDS42"
```

```
anno
```

```
       structureId chainId macromoleculeType chainLength experimentalTechnique
1AKE_A        1AKE       A           Protein         214                 X-ray
6S36_A        6S36       A           Protein         214                 X-ray
6RZE_A        6RZE       A           Protein         214                 X-ray
3HPR_A        3HPR       A           Protein         214                 X-ray
1E4V_A        1E4V       A           Protein         214                 X-ray
5EJE_A        5EJE       A           Protein         214                 X-ray
1E4Y_A        1E4Y       A           Protein         214                 X-ray
3X2S_A        3X2S       A           Protein         214                 X-ray
6HAP_A        6HAP       A           Protein         214                 X-ray
       resolution       scopDomain                                      pfam
1AKE_A       2.00 Adenylate kinase Adenylate kinase, active site lid (ADK_lid)
6S36_A       1.60             <NA>                      Adenylate kinase (ADK)
6RZE_A       1.69             <NA>                      Adenylate kinase (ADK)
3HPR_A       2.00             <NA> Adenylate kinase, active site lid (ADK_lid)
1E4V_A       1.85 Adenylate kinase                      Adenylate kinase (ADK)
5EJE_A       1.90             <NA> Adenylate kinase, active site lid (ADK_lid)
1E4Y_A       1.85 Adenylate kinase Adenylate kinase, active site lid (ADK_lid)
3X2S_A       2.80             <NA>                                         <NA>
6HAP_A       2.70             <NA>                      Adenylate kinase (ADK)
               ligandId
1AKE_A              AP5
6S36_A CL (3),NA,MG (2)
6RZE_A    NA (3),CL (2)
3HPR_A              AP5
1E4V_A              AP5
5EJE_A           AP5,CO
1E4Y_A              AP5
3X2S_A   JPY (2),AP5,MG
6HAP_A              AP5
                                                                   ligandName
1AKE_A                                                BIS(ADENOSINE)-5'-PENTAPHOSPHATE
6S36_A                                   CHLORIDE ION (3),SODIUM ION,MAGNESIUM ION (2)
6RZE_A                                                SODIUM ION (3),CHLORIDE ION (2)
3HPR_A                                                BIS(ADENOSINE)-5'-PENTAPHOSPHATE
```

```
1E4V_A                                                BIS(ADENOSINE)-5'-PENTAPHOSPHATE
5EJE_A                                 BIS(ADENOSINE)-5'-PENTAPHOSPHATE,COBALT (II) ION
1E4Y_A                                                BIS(ADENOSINE)-5'-PENTAPHOSPHATE
3X2S_A N-(pyren-1-ylmethyl)acetamide (2),BIS(ADENOSINE)-5'-PENTAPHOSPHATE,MAGNESIUM ION
6HAP_A                                                BIS(ADENOSINE)-5'-PENTAPHOSPHATE
                                          source
1AKE_A                         Escherichia coli
6S36_A                         Escherichia coli
6RZE_A                         Escherichia coli
3HPR_A                    Escherichia coli K-12
1E4V_A                         Escherichia coli
5EJE_A   Escherichia coli O139:H28 str. E24377A
1E4Y_A                         Escherichia coli
3X2S_A Escherichia coli str. K-12 substr. MDS42
6HAP_A   Escherichia coli O139:H28 str. E24377A

1AKE_A STRUCTURE OF THE COMPLEX BETWEEN ADENYLATE KINASE FROM ESCHERICHIA COLI AND THE INHIBI
6S36_A
6RZE_A
3HPR_A
1E4V_A
5EJE_A                                                                                  Crys
1E4Y_A
3X2S_A
6HAP_A
                                                 citation rObserved  rFree
1AKE_A             Muller, C.W., et al. J Mol Biology (1992)    0.1960     NA
6S36_A                 Rogne, P., et al. Biochemistry (2019)    0.1632 0.2356
6RZE_A                 Rogne, P., et al. Biochemistry (2019)    0.1865 0.2350
3HPR_A Schrank, T.P., et al. Proc Natl Acad Sci U S A (2009)    0.2100 0.2432
1E4V_A                  Muller, C.W., et al. Proteins (1993)    0.1960     NA
5EJE_A Kovermann, M., et al. Proc Natl Acad Sci U S A (2017)    0.1889 0.2358
1E4Y_A                  Muller, C.W., et al. Proteins (1993)    0.1780     NA
3X2S_A               Fujii, A., et al. Bioconjug Chem (2015)    0.2070 0.2560
6HAP_A              Kantaev, R., et al. J Phys Chem B (2018)    0.2263 0.2776
        rWork spaceGroup
1AKE_A 0.1960  P 21 2 21
6S36_A 0.1594    C 1 2 1
6RZE_A 0.1819    C 1 2 1
3HPR_A 0.2062  P 21 21 2
1E4V_A 0.1960  P 21 2 21
5EJE_A 0.1863  P 21 2 21
1E4Y_A 0.1780   P 1 21 1
```

```
3X2S_A 0.2070 P 21 21 21
6HAP_A 0.2237    I 2 2 2
```

##Principal component analysis

```
pc.xray <- pca(pdbs)
plot(pc.xray)
```

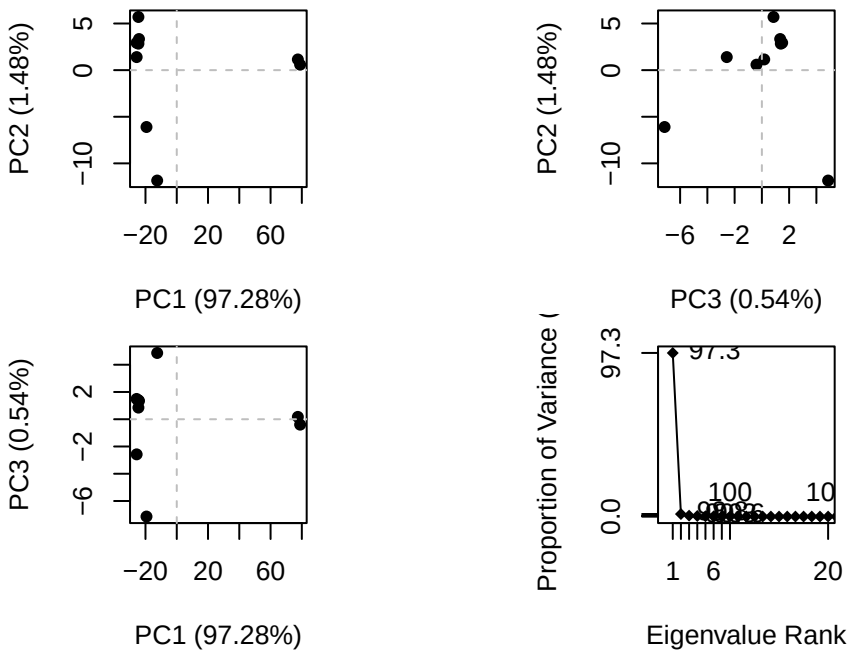


Figure 9: PCA of adenylate kinase structures. Each dot represents one structure. PC1 explains most of the variance, showing that the main structural differences among adenylate kinase conformations occur along this first principal component.

##RMSD + clustering + color the PCA plot

```
rd <- rmsd(pdbs)
```

```
Warning in rmsd(pdbs): No indices provided, using the 214 non NA positions
```

```
hc.rd <- hclust(dist(rd))
grps.rd <- cutree(hc.rd, k=3)

plot(pc.xray, 1:2, col="purple", bg=grps.rd, pch=21, cex=1)
```

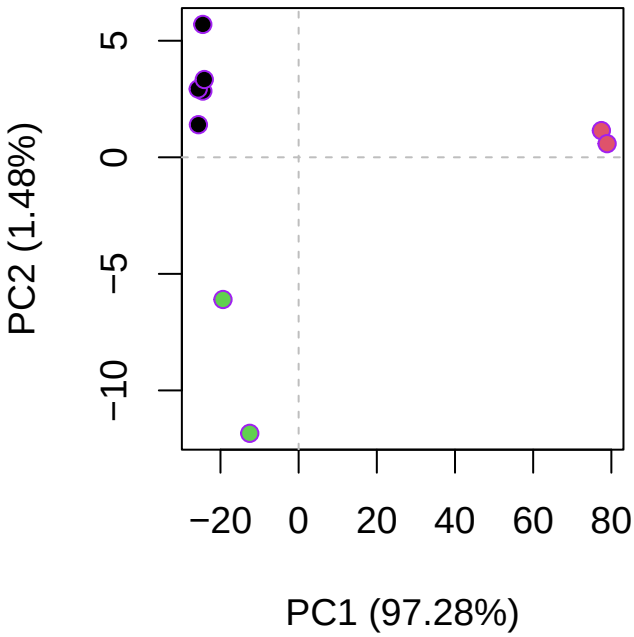


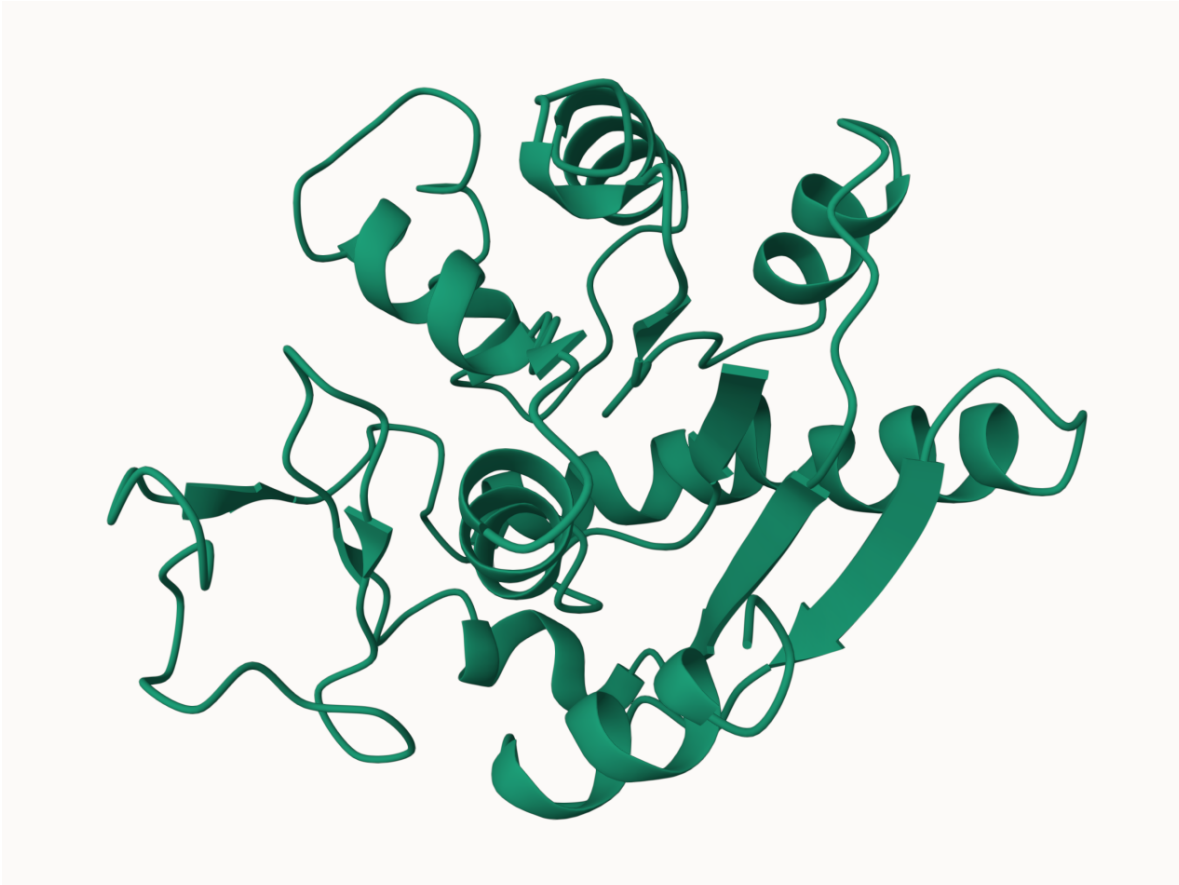
Figure 10: Projection of Adenylate kinase X-ray structures. Each dot represents one PDB structure.

Finaly let's make a wee movie of the major "motion" or strucutral differences in the dataset - we call this a

```
mktrj(pc.xray, file="pc_1.pdb")
```

We can make a visialization of the major conformational differences (i.e. large scale strucutre change)

```
pc1 <- mktrj(pc.xray, pc=1, file="pc_1.pdb")
```



We can also plot our main PCA results with ggplot:

```
library(ggplot2)
```

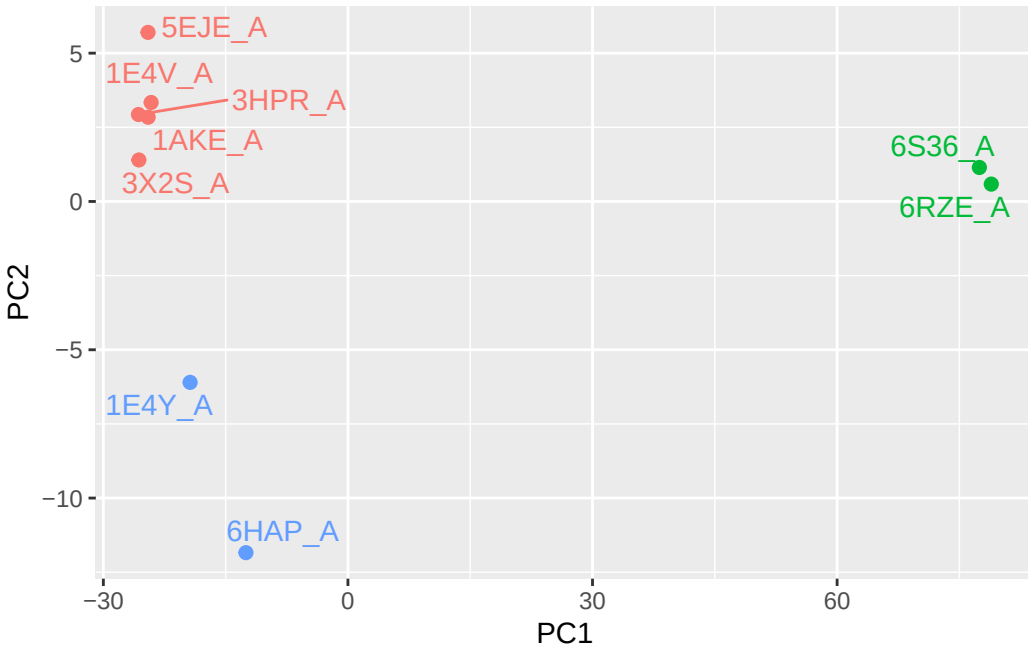
```
Warning: package 'ggplot2' was built under R version 4.5.2
```

```
library(ggrepel)

df <- data.frame(PC1=pc.xray$z[,1],
                 PC2=pc.xray$z[,2],
                 col=as.factor(grps.rd),
                 ids=ids)

p <- ggplot(df) +
  aes(PC1, PC2, col=col, label=ids) +
  geom_point(size=2) +
```

```
  geom_text_repel(max.overlaps = 20) +
  theme(legend.position = "none")
p
```



##Normal mode analysis

```
modes <- nma(pdbs)
```

```
Details of Scheduled Calculation:
  ... 9 input structures
  ... storing 636 eigenvectors for each structure
  ... dimension of x$U.subspace: ( 642x636x9 )
  ... coordinate superposition prior to NM calculation
  ... aligned eigenvectors (gap containing positions removed)
  ... estimated memory usage of final 'eNMA' object: 28.1 Mb


  |
  |                                                                      |   0%
  |
  |========                                                              |  11%
```

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  |
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  |=====================================================                 |  78%
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  |================================================================      |  89%
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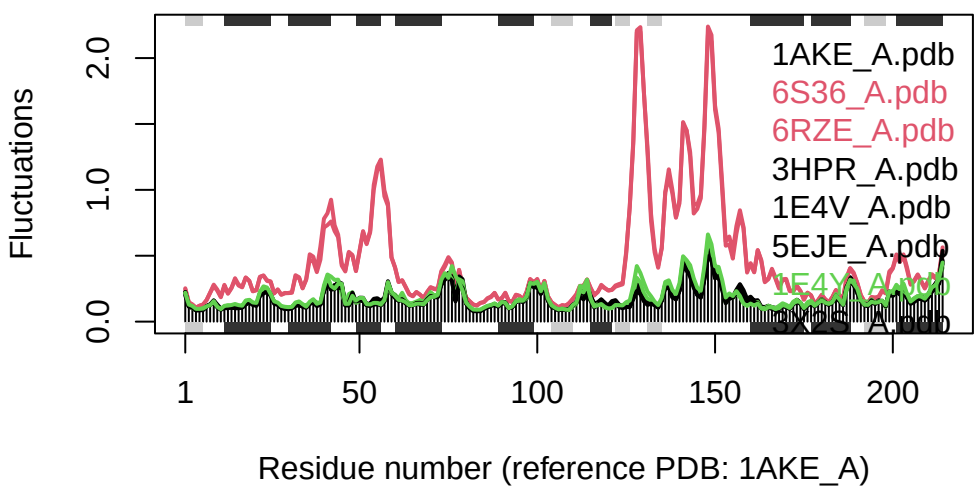
  |======================================================================| 100%
```

```
plot(modes, pdbs, col=grps.rd)
```

```
Extracting SSE from pdbs$sse attribute
```



Q14. What do you note about this plot? Are the black and colored lines similar or different? Where do you think they differ most and why?

The black and colored lines are mostly similar overall, but they are most different in the big peak region around 130–160 residues and also a smaller difference around 40–60 residues. That region shows the highest fluctuations, meaning it's the most flexible part and likely corresponds to mobile nucleotide binding.